SMARTBRIDGE & SMART INTERNZ PROJECT

LegalEase

AI-Powered Legal Document Generator bridging accessibility and legal professionalism through Generative AI, FastAPI, and Streamlit.

Project Architecture & Overview

Building a robust, end-to-end full-stack SaaS solution for instant, customized legal documentation.

Core Concept & Real-World Scenarios

Democratizing Legal Content

LegalEase leverages state-of-the-art Generative AI to simplify legal document creation. Entrepreneurs, freelancers, and individuals can generate reliable contracts, lease agreements, and NDAs without a legal background.

Practical Use Cases

Startup Founders: Tailor employment contracts with role details and logos.
Freelancers: Instant NDAs specifying confidentiality scope.
Landlords: Formatted residential lease agreements with auto term tables.

System Workflow & Technology Stack

- **Frontend UI:** Streamlit powers user inputs, dynamic document preview, and multi-format exports.
- **Backend API:** FastAPI (main.py, routes.py) handles secure JSON payload requests and processing.
- **AI Core Engine:** Google Gemini 1.5 Pro generates formal legal structure with precise clauses.
- **Export Modules:** python-docx and FPDF convert text into styled Word and branded PDF documents.



Core Development Milestones

Step-by-step implementation from model selection to deployment and verification.

Milestones 1 & 2: Model & Core Features

Model Selection

Selected Gemini 1.5 Pro for its long context window, fast generation speed, and superior legal-domain reasoning capabilities for complex clauses.

Core Generation

Developed FastAPI backend endpoints (/generate) and text formatting engines to sanitize output and format professional documents automatically.

Editable Previews

Enabled interactive Streamlit text areas for inline editing prior to final document export in .TXT, .DOCX, or .PDF formats.

Backend API Logic & Implementation

Module / File	Primary Responsibility	Key Framework / Library
`main.py`	Initializes FastAPI app, sets up health-check routes, and includes API routers.	FastAPI, Uvicorn
`routes.py`	Defines Pydantic models (DocumentRequest) and processes POST requests.	Pydantic, APIRouter
`gemini_generator.py`	Configures Google GenAI SDK and prompts Gemini 1.5 Pro model.	google-generativeai
`app.py`	Streamlit user interface for inputs, styling, and multi-format file downloads.	Streamlit, Requests

Frontend & Streamlit UI

Designed an intuitive, responsive Streamlit web interface featuring center-aligned branding logos, text input fields, and structured text areas for parties and terms.

Users can instantly generate agreements, view styled dark-theme HTML previews, edit text inline, and download polished deliverables complete with logos and footers.



Deployment & Local Verification

- **Environment Setup:** Isolated virtual environment created with python-dotenv for secure Gemini API key management.
- **Dependency Management:** Installed production-ready packages including FastAPI, Uvicorn, Streamlit, python-docx, FPDF, and Pillow.
- **Backend Launch:** Started Uvicorn server successfully at `http://127.0.0.1:8000` with auto-reload enabled.
- **Frontend Execution:** Launched Streamlit UI at `http://localhost:8501` for seamless end-to-end testing across diverse input combinations.

Conclusion & Future Enhancements

Legal Empowerment

Democratizes legal access by transforming complex legal language into clear, understandable documents for non-legal professionals.

Scalable Architecture

Demonstrates efficient AI model integration, secure data handling, and responsive SaaS design principles ready for cloud scaling.

Future Roadmap

Plans include deeper contract analysis, legal database integration, multilingual support, and personalized recommendations.

Questions?

Thank you for exploring LegalEase. Ready to revolutionize legal document generation.

SmartBridge & Smart Internz Project | LegalEase AI

Image Sources



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